Physics 468/668: Statistical Physics **Spring 2019**

Lecture 18: Condensation of a Van der Waals gas

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Various physical phenomena to which the formalism of statistical mechanics has been applied may, in general, be divided into two categories. In the first category, the microscopic constituents of the given system are, or can be regarded as, practically noninteracting, including specific heats of gases, the specific heats of solids (Section 7.4), chemical reactions and equilibrium constants (Section 6.6), the condensation of an ideal Bose gas (Sections 7.1 and 7.2). In the case of solids, the interatomic interaction does, in fact, play an important physical role; however, since the actual positions of the atoms, over a substantial range of temperatures, do not depart significantly from their mean values, we can rewrite our problem in terms of the so-called normal coordinates and treat the given solid as an assembly of practically noninteracting harmonic oscillators.

Phenomena belonging to the second category, however, present a very different situation. In most cases, one encounters analytic discontinuities or singularities in the thermodynamic functions of the given system which, in turn, correspond to the occurrence of various kinds of phase transitions. Notable examples of phenomena belonging to this category are the condensation of gases, the melting of solids, phenomena associated with the coexistence of phases (especially in the neighborhood of a critical point), the behavior of mixtures and solutions (including the onset of phase separation), the orderdisorder transitions in alloys, the transition from a normal to a superconducting material, and so on.

Mathematical problems associated with the study of cooperative phenomena are quite formidable.1 To facilitate calculations, one is forced to introduce models in which the interactions are considerably simplified, yet retain characteristics that are essential to the cooperative aspect of the problem. One then hopes that a theoretical study of these simplified models, which still involves serious difficulties of analysis, will reproduce the most basic features of the phenomena exhibited by actual physical systems.

18.1 van der Waals gas

Now we start with the following equation of state,

$$P = \frac{RT}{v-b} - \frac{a}{v^2} \tag{18.1}$$

The isotherms are shown in Figure 18.2. As shown at high temperature, the slope of $\frac{\partial P}{\partial v}$-curve is never positive. This can be understood as follows. **For a system is under equilibrium, if its volume expands, the pressure has to be reduced. Then the external environment will force the system return to the original states. Otherwise, the system is not stable.** For a rigorous mathematical proof, you can check the 2nd derivative of $\Delta^2 U > 0$ (i.e., U must be at minimum).

However, at lower temperatures there can exist a region in which the slope is zero. The existence of such regions corresponds to the coexistence of two or more phases. In other words, this indicates an onset of phase transition. Regions with positive slope should never appear. The boundary of such forbidden region can be graphically analyzed by *Maxwell construction*. The physical basis of the Maxwell construction can be seen with the help of $\mu(T, P)$. When liquid and gas reach the equilibrium,

$$\mu_{\text{liquid}} = \mu_{\text{gas}} \tag{18.2}$$

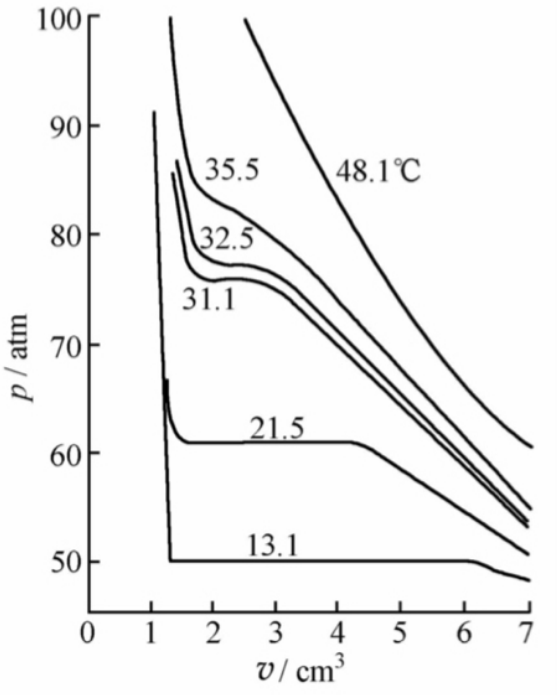


Figure 18.1: The experimentally measured CO_2 isotherms.

In the following figure, from point 1 to 3, we follow that

$$\begin{aligned} \mu_3 &= \mu_1 + \int_{P_1}^{P_3} \frac{\partial \mu}{\partial P} dP \\ &= \mu_1 + \int_{P_1}^{P_3} \frac{V}{N} dP \end{aligned} \tag{18.3}$$

To ensure that $\mu_1 = \mu_3$ from the theoretical isotherm, it require that the quantity VdP integrated along the isotherm from state 1 to state 3 much vanish. That means the area from 1-2 to 2-3 should be canceled out.

When you gradually increase the temperature from low to high, the size of "forbidden region" gradually shrinks and vanish. So there is a critical temperature T_c. It corresponds to the point $(\partial^2 P/\partial v^2)_T = 0$. Using Eq. 18.1, we obtain the coordinates of the critical point,

$$\begin{aligned} \frac{\partial P}{\partial v} &= -\frac{RT}{(v-b)^2} + \frac{2a}{v^3} = 0 \\ \frac{\partial^2 P}{\partial v^2} &= \frac{2RT}{(v-b)^3} - \frac{6a}{v^4} = 0 \end{aligned}$$

$$P_c = \frac{a}{27b^2}, \quad v_c = 3b, \quad T_c = \frac{8a}{27bR} \tag{18.4}$$

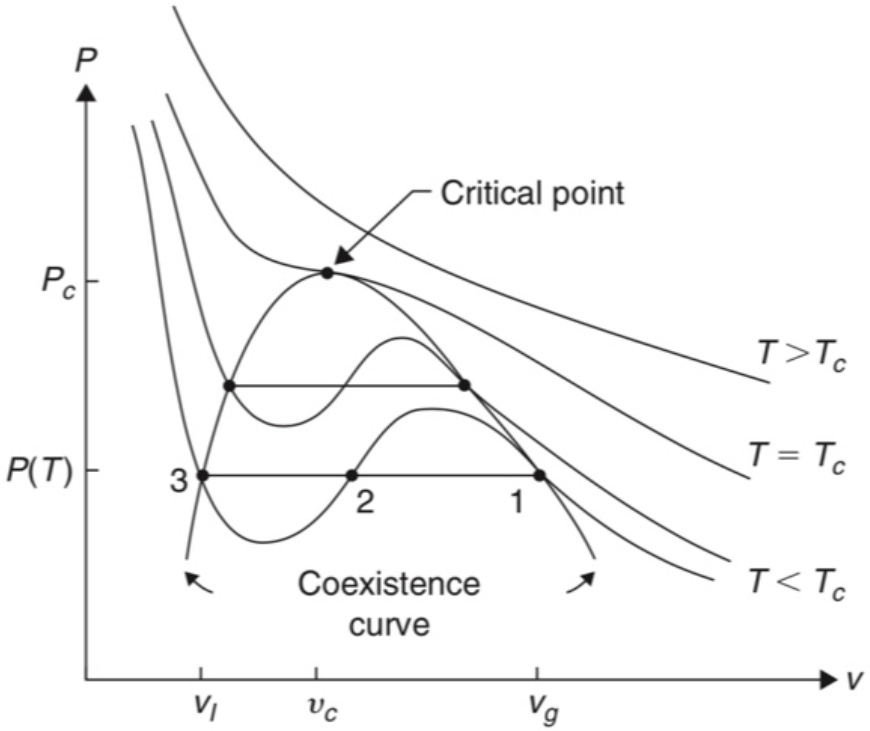


Figure 18.2: The isotherms of a van der Waals system; those for $T < T_c$ are "corrected with the help of the Maxwell construction, thus leading to the coexistence curve at the top of which sits the critical point.

We thus find that P_c, v_c, and T_c vary from system to system (through the interaction parameters a and b). However, the quantity K,

$$K = RT_c/P_c v_c = 8/3 = 2.66666 \tag{18.5}$$

has a common, universal value.

$$\kappa_T = -\frac{1}{V}\left(\frac{\partial V}{\partial P}\right)_T = -\frac{1}{V(\partial P/\partial V)} \tag{18.6}$$

At the critical point,

$$\begin{aligned}\frac{\partial P}{\partial V}|_{v=v_c} &= -\frac{RT}{(v_c-b)^2} + \frac{2a}{v_c^3} \\ &= -\frac{RT}{4b^2} + \frac{2a}{27b^3} \\ &= \frac{R}{4b^2}\left(\frac{8a}{27Rb} - T\right) \\ &= \frac{R}{4b^2}\left(T_c - T\right)\end{aligned} \tag{18.7}$$

Therefore,

$$\kappa_T \propto (T - T_c)^{-1} \tag{18.8}$$

$$P = P_c + \frac{\partial P}{\partial \rho}(\rho - \rho_c) + \frac{1}{2}\frac{\partial^2 P}{\partial \rho^2}(\rho - \rho_c)^2 + \cdots \tag{18.9}$$

At $P = P_c$

$$\begin{aligned} \frac{\partial P}{\partial \rho} &= \frac{\partial P}{\partial V}\frac{\partial V}{\partial \rho} = 0 \\ \frac{\partial^2 P}{\partial \rho^2} &= \frac{\partial^2 P}{\partial V^2}\left(\frac{\partial V}{\partial \rho}\right)^2 + \frac{\partial P}{\partial V}\frac{\partial^2 V}{\partial \rho^2} = 0 \end{aligned} \tag{18.10}$$

So the first nonzero term should be the third order,

$$(P - P_c) \propto (\rho - \rho_c)^{-3} \tag{18.11}$$

In general, one can prove the followings

$$C_V = (T - T_c)^{-\alpha} \quad (\alpha = 0, \text{Experiment: 0.1-0.2}) \tag{18.12}$$

$$\rho_l - \rho_g = (T - T_c)^{-\beta} \quad (\beta = 0.5, \text{Experiment: 0.32-0.35}) \tag{18.13}$$

$$\kappa = -\frac{1}{V}\left(\frac{\partial V}{\partial P}\right) = (T - T_c)^{-\gamma} \quad (\gamma = 1, \text{Experiment: 1.2-1.3}) \tag{18.14}$$

$$P - P_c = (\rho - \rho_c)^{\delta} \quad (\delta = 3, \text{Experiment: 4.6-5.0}) \tag{18.15}$$

The above four equations illustrate the nature of the critical behavior displayed by a van der Waals system undergoing the gasliquid transition. While it differs in several important respects from the critical behavior of real physical systems, it shows up again and again in studies pertaining to other critical phenomena that have apparently nothing to do with the gas-liquid phase transition.